Exercise Set 13: Symplectic Geometry and Hamiltonian Mechanics

A symplectic manifold (M, ω) is a smooth manifold equipped with a closed, non-degenerate 2-form ω. This single object the symplectic form encodes the entire structure of classical mechanics: positions, momenta, Poisson brackets, conservation laws, and the equations of motion. The Kepler problem, the oldest and deepest problem in physics, finds its natural and complete solution in this language. In the final section of this set, we look back at the entire journey from Set 1 to Set 13 and see what each new layer of structure gave us.

Recall of Definitions and Examples

Symplectic manifolds

R1. Write the definition of a **symplectic manifold** (M, ω). What two conditions must $\omega \in \Omega^2(M)$ satisfy?

(a) **Closed**: $d\omega = ?$

(b) **Non-degenerate**: for every $p \in M$ and nonzero $v \in T_pM$, there exists $w \in T_pM$ such that $\omega(v, w) \neq 0$.

What does non-degeneracy imply about $\dim M$? Why can odd-dimensional manifolds not be symplectic?

R2. The fundamental example: the **cotangent bundle** T^*Q of any smooth manifold Q (the configuration space).

(a) Write the **Liouville 1-form** (tautological form) $\alpha \in \Omega^1(T^*Q)$ in local coordinates (q^i, p_i):

$$\alpha = \sum_i p_i \, dq^i.$$

(b) Write the **canonical symplectic form**: $\omega = -d\alpha = ?$

(c) Verify $d\omega = 0$ and non-degeneracy of ω.

R3. State the **Darboux theorem**: every symplectic manifold (M, ω) of dimension $2n$ locally looks like $(T^*\mathbb{R}^n, \omega_0)$ where $\omega_0 = \sum_{i=1}^n dq^i \wedge dp_i$.

What does this say about symplectic geometry vs Riemannian geometry? *(In Riemannian geometry, the curvature is a local invariant. In symplectic geometry, there is no local invariant all symplectic manifolds are locally the same.)*

R4. Give examples of symplectic manifolds:

(a) $(\mathbb{R}^{2n}, \omega_0)$ with $\omega_0 = \sum dq^i \wedge dp_i$

(b) (T^*S^1, ω) the phase space of the pendulum

(c) (T^*S^2, ω) the phase space of a particle on S^2

(d) $(S^2, \sin\theta\, d\theta \wedge d\varphi)$ the **coadjoint orbit** of $SO(3)$, the phase space of a spinning top

(e) $(T^2, d\theta_1 \wedge d\theta_2)$ but is this the phase space of something?

Hamiltonian mechanics

R5. Write the definition of the **Hamiltonian vector field** $X_H \in \mathfrak{X}(M)$ associated to $H \in C^\infty(M)$:

$$\iota_{X_H}\omega = -dH \qquad \text{(i.e. } \omega(X_H, \cdot) = -dH\text{)}.$$

Why does non-degeneracy of ω guarantee X_H exists and is unique? In coordinates (q^i, p_i), show $X_H = \sum_i \left(\frac{\partial H}{\partial p_i}\partial_{q^i} - \frac{\partial H}{\partial q^i}\partial_{p_i}\right)$.

R6. Write the definition of the **Poisson bracket**:

$$\{f, g\} = \omega(X_f, X_g) = \sum_i \left(\frac{\partial f}{\partial q^i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q^i}\right).$$

State:

(a) $\{q^i, p_j\} = ?$

(b) $\{q^i, q^j\} = ?$

(c) $\{p_i, p_j\} = ?$

(d) $\{H, f\} = 0$ means: f is **conserved** along the flow of X_H.

R7. State **Liouville's theorem**: the Hamiltonian flow preserves ω:

$$\mathcal{L}_{X_H}\omega = 0.$$

Prove it using Cartan's formula: $\mathcal{L}_{X_H}\omega = d(\iota_{X_H}\omega) + \iota_{X_H}(d\omega) = d(-dH) + 0 = 0$.

What does this say about phase space volume?

R8. State the **Jacobi identity** for the Poisson bracket:

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0.$$

What algebraic structure does $(C^\infty(M), \{\cdot, \cdot\})$ form? How does this connect to the Lie algebra of vector fields?

Theory

T1. (Hamiltonian flow preserves ω Liouville's theorem.) Give the full proof of $\mathcal{L}_{X_H}\omega = 0$ using Cartan's formula. Then prove the consequence: the Hamiltonian flow ϕ_t preserves the Liouville volume form $\omega^n = \omega\wedge\cdots\wedge\omega$ (n times). In coordinates, this says $\det \frac{\partial(q(t),p(t))}{\partial(q(0),p(0))} = 1$ the flow is volume-preserving.

T2. (Conservation laws and Poisson brackets.) Let $f \in C^\infty(M)$. Show:

(a) $\frac{d}{dt} f(\phi_t(x)) = \{H, f\}(\phi_t(x))$ where ϕ_t is the Hamiltonian flow.

(b) f is conserved along orbits iff $\{H, f\} = 0$.

(c) If $\{H, f\} = 0$ and $\{H, g\} = 0$, show $\{H, \{f, g\}\} = 0$ using the Jacobi identity. Conclude: the conserved quantities form a Lie algebra under the Poisson bracket.

T3. (The symplectic form on S^2 as a coadjoint orbit.)

The sphere S^2 with $\omega = \sin\theta\, d\theta \wedge d\varphi$ is a symplectic manifold of dimension 2.

(a) Verify $d\omega = 0$.

(b) Verify ω is non-degenerate on $S^2 \setminus \{N, S\}$.

(c) The functions $J_x = \sin\theta\cos\varphi$, $J_y = \sin\theta\sin\varphi$, $J_z = \cos\theta$ on S^2 satisfy the Poisson bracket relations (with respect to ω):

$$\{J_x, J_y\} = J_z, \qquad \{J_y, J_z\} = J_x, \qquad \{J_z, J_x\} = J_y.$$

This is the Lie algebra $\mathfrak{so}(3)$ appearing as Poisson brackets the sphere is the **coadjoint orbit** of $SO(3)$. Verify the first relation $\{J_x, J_y\} = J_z$ explicitly.

T4. (Noether's theorem in symplectic language.) Let X be a vector field on (M, ω) generating a flow that preserves H: $\mathcal{L}_X H = X(H) = 0$.

(a) If $X = X_f$ is a Hamiltonian vector field (so $\iota_{X_f}\omega = -df$), show $X(H) = \{f, H\} = -\{H, f\}$.

(b) Conclude: X preserves H iff $\{H, f\} = 0$ iff f is conserved.

(c) Compare with the Noether theorem from Set 7 (P2) where we used $\mathcal{L}_X L = 0$. Write one sentence on what the symplectic language adds: what is the role of ω here that the Lagrangian language did not have?

Computation

C1. (Hamiltonian flow on $\mathbb{R}^{2n}$.)

(a) For $H = \frac{p^2}{2m} + V(q)$ (particle in a potential), write the Hamiltonian vector field X_H and show the flow equations are Newton's law $m\ddot{q} = -V'(q)$.

(b) For $H = \frac{1}{2}(p^2 + \omega^2 q^2)$ (harmonic oscillator), write X_H and solve the flow equations explicitly. Describe the orbits in phase space.

(c) Verify Liouville: compute $\mathcal{L}_{X_H}(dp \wedge dq)$ for the harmonic oscillator and show it vanishes.

C2. (Poisson brackets of angular momentum.)

On $T^*\mathbb{R}^3$ with $\omega = \sum_{i=1}^{3} dp_i \wedge dq^i$, define the angular momentum components:

$$J_1 = q^2 p_3 - q^3 p_2, \quad J_2 = q^3 p_1 - q^1 p_3, \quad J_3 = q^1 p_2 - q^2 p_1.$$

(a) Compute $\{J_1, J_2\}$ using the canonical Poisson brackets $\{q^i, p_j\} = \delta^i_j$. Show $\{J_1, J_2\} = -J_3$ (or $+J_3$ depending on convention).
This is the Lie algebra $\mathfrak{so}(3)$ appearing in the Poisson bracket connecting to Set 11.

(b) Show $\{J_i, J_j\} = -\epsilon_{ijk} J_k$.

(c) Let $J^2 = J_1^2 + J_2^2 + J_3^2$. Show $\{J_i, J^2\} = 0$ for all i.

(d) For $H = \frac{|\mathbf{p}|^2}{2m} + V(|\mathbf{q}|)$ (central force), show $\{H, J_i\} = 0$. What does this say physically?

C3. (The pendulum in symplectic language.)

The pendulum has configuration space S^1 and phase space $T^*S^1 \cong S^1 \times \mathbb{R}$ with coordinates (θ, p_θ) and symplectic form $\omega = dp_\theta \wedge d\theta$.

(a) Write the Hamiltonian $H = \frac{p_\theta^2}{2} - \cos\theta$. (Unit mass and length.)

(b) Compute X_H using $\iota_{X_H}\omega = -dH$. Verify you recover the equations from Set 7 (P1).

(c) Show $\{H, p_\theta\} \neq 0$ the momentum p_θ is not conserved. Is energy conserved?

(d) The equilibria are where $X_H = 0$. Find them. Compare with Set 7 (P1c).

(e) Compute $\mathcal{L}_{X_H}\omega$ and verify it vanishes.

C4. (The Liouville 1-form in polar coordinates.)

This is the setup for the Kepler problem (Nawata HW6 P1 Part 3).

On $T^*(\mathbb{R}^3 \setminus \{0\})$ in polar coordinates:

$$x^1 = r\sin\theta\cos\varphi, \quad x^2 = r\sin\theta\sin\varphi, \quad x^3 = r\cos\theta.$$

(a) The Liouville 1-form in Cartesian coordinates is $\alpha = \sum_i p_i \, dq^i$. Rewrite in polar coordinates: show

$$\alpha = p_r \, dr + p_\theta \, d\theta + p_\varphi \, d\varphi$$

where p_r, p_θ, p_φ are the conjugate momenta.

(b) Write $\omega = -d\alpha$ in polar coordinates.

(c) Rewrite the conserved quantities H, J_3, and J^2 in polar coordinates and in terms of $(p_r, p_\theta, p_\varphi)$.
Hint: $J_3 = p_\varphi$ in these coordinates.

Physics Part A: The Kepler Problem

The Kepler problem a planet orbiting the Sun is the oldest and most celebrated integrable system in physics. In symplectic language, it is the Hamiltonian system on $T^*(\mathbb{R}^3 \setminus \{0\})$ with $H = \frac{|\mathbf{p}|^2}{2} - \frac{k}{r}$. Three conserved quantities H, J_3, and J^2 make it completely integrable via the Liouville theorem. Following Nawata HW6, we derive Kepler's three laws from the symplectic structure.

A1. (Angular momentum is conserved Nawata HW6 P1a.)

For $H = \frac{|\mathbf{p}|^2}{2} - \frac{k}{r}$:

(a) Show $\{H, J_i\} = 0$ for all $i = 1, 2, 3$. Conclude: all three components of angular momentum are conserved.

(b) The three quantities H, J_3, J^2 are in involution: show $\{H, J_3\} = 0$, $\{H, J^2\} = 0$, $\{J_3, J^2\} = 0$.

(c) Since we have 3 independent conserved quantities in involution on a 6-dimensional phase space (and $3 = 6/2$), the system is **completely integrable** by the Liouville-Arnold theorem.

A2. (Kepler's second law conservation of angular momentum.)

(a) Without loss of generality, rotate coordinates so $\vec{J} = (0, 0, J_3)$, i.e. set $\theta = \pi/2$. In this plane, $J_3 = r^2\dot{\varphi}$.

(b) **Kepler's second law**: the areal velocity $\frac{dA}{dt} = \frac{1}{2}r^2\dot{\varphi} = \frac{J_3}{2}$ is constant. Show this follows directly from $\{H, J_3\} = 0$.

(c) The swept area in time dt is $dA = \frac{1}{2}r^2 \, d\varphi$. Equal areas in equal times is exactly $J_3 = \text{const}$.

A3. (Hamilton-Jacobi and separation of variables Nawata HW6 P1d.)

Setting $\theta = \pi/2$ (motion in the equatorial plane): the Liouville 1-form becomes $\alpha = p_r\, dr + J_3\, d\varphi$, and integrating gives the **Hamilton's principal function**:

$$S = \int \alpha = \pm \int^r dr \sqrt{2\left(H + \frac{k}{r}\right) - \frac{J_3^2}{r^2}} + J_3\varphi.$$

(a) The equations of motion for the **angle variables** are:

$$\psi_H = \frac{\partial S}{\partial H} = \text{const}, \qquad \psi_J = \frac{\partial S}{\partial J_3} = \text{const}.$$

Compute $\psi_H = \frac{\partial S}{\partial H}$. Show it gives t as a function of r the radial period.

(b) Compute $\psi_J = \frac{\partial S}{\partial J_3}$. Show it gives φ as a function of r the shape of the orbit.

A4. (Kepler's first law: elliptical orbits Nawata HW6 P1e.)

For $V(r) = -k/r$:

(a) From $\psi_J = \text{const}$, derive the orbit equation:

$$r(\varphi) = \frac{\ell}{1 + e\cos\varphi}$$

where $\ell = J_3^2/k$ is the **semi-latus rectum** and $e = \sqrt{1 + \frac{2HJ_3^2}{k^2}}$ is the **eccentricity**.

(b) Show: for $E < 0$ (bound orbit), $e < 1$ and the orbit is an ellipse with the Sun at one focus. This is Kepler's first law.

(c) Identify the semi-major axis a in terms of H and k: show $a = -k/(2H)$.

A5. (Kepler's third law: $T^2 \propto a^3$ Nawata HW6 P1e.)

(a) From $\psi_H = \text{const}$, compute the orbital period T by integrating $dt = dr/\dot{r}$ over one full orbit. Show:

$$T = \frac{2\pi}{\sqrt{k}}\, a^{3/2}.$$

(b) This is Kepler's third law: $T^2 = \frac{4\pi^2}{k} a^3$. The constant of proportionality $\frac{4\pi^2}{k}$ is the same for all planets (only the mass of the Sun appears, not the planet).

(c) For the Earth-Sun system, $k = GM_\odot$. Verify the formula gives $T \approx 1$ year for $a \approx 1$ AU.

Physics Part B: Building Physics from Scratch

We started in Set 1 with a set X and a function $f : X \to X$. We could describe orbits, fixed points, and periodic points. We could not say anything about convergence, speed, or

equations of motion. In thirteen sets we have added structure layer by layer, and at each step we gained something genuinely new. Here we collect the complete picture and state precisely what each new structure provides and what it cannot.

B1. (The complete ladder of structures.)

Fill in the following table. For each row, state: (i) what new structure was added, (ii) what became possible that was not possible before, (iii) one fundamental limitation what you still cannot do.

Structure	**What it gives**	**What is still missing**
Set (Sets 1–2)	Orbits, fixed points, periodic points	No notion of closeness
Topology (Sets 3–4)	Continuity, convergence, connectedness, compactness	Limits not unique; no speed
Hausdorff (Set 4)	Unique limits	No smooth structure
Smooth manifold (Sets 5–6)	Velocity, differentiation, ODE on M	No way to measure length or angle
Tangent bundle (Sets 7)	Equations of motion as vector fields, Noether's theorem	No pairing between positions and momenta
Differential forms (Sets 8–9)	Integration, Stokes' theorem, gauge fields	No canonical pairing between TM and T^*M
De Rham cohomology (Set 10)	Topological invariants from analysis, Witten index	No group structure on the symmetries
Lie groups (Set 11)	Continuous symmetries, Lie algebra, spin	No phase space structure
Fiber bundles (Set 12)	Gauge fields, connections, curvature, monopoles	No canonical structure on T^*M
Symplectic form (Set 13)	Phase space, Poisson brackets, Hamiltonian flow	No metric, no curvature

B2. (What is genuinely new in symplectic geometry.)

The symplectic form ω is unlike all previous structures:

(a) A Riemannian metric g is a symmetric non-degenerate bilinear form. The symplectic form ω is antisymmetric and non-degenerate. What does antisymmetry imply for $\omega(v, v)$? Why does this mean there are no preferred directions in phase space?

(b) A Riemannian metric identifies $TM \cong T^*M$ via $g(v, \cdot) : TM \to T^*M$. The symplectic form identifies $TM \cong T^*M$ via $\omega(v, \cdot) : TM \to T^*M$. For a Hamiltonian system: the gradient $dH \in T^*M$ is mapped to the Hamiltonian vector field $X_H \in TM$ via $\iota_{X_H}\omega = -dH$. This is the fundamental duality between observables and flows. Explain why this duality does not exist in Riemannian geometry.

(c) The Darboux theorem says all symplectic manifolds are locally equivalent there is no local symplectic invariant (no "symplectic curvature"). This is the opposite of Riemannian geometry where curvature is a local invariant. Why is this physically natural for phase space?
(Hint: all classical mechanics is locally the same positions and momenta always satisfy $\{q^i, p_j\} = \delta^i_j$, regardless of the shape of configuration space.)

(d) The conservation law in symplectic language is: f is conserved iff $\{H, f\} = 0$ iff X_f preserves H. In Riemannian language (Set 7): f is conserved iff $\mathcal{L}_{X_f} L = 0$. What does the symplectic language add?
(Answer: the Poisson bracket $\{f, g\}$ is itself a conserved quantity when f and g are the conserved quantities form a Lie algebra. The Riemannian/Lagrangian language does not have this structure automatically.)

B3. (The running problem complete solution.)

In Set 2 we started with $f : X \to X$ on a finite set X and asked: what are the orbits?

Now the full Kepler system lives on the symplectic manifold $T^*(\mathbb{R}^3 \setminus \{0\})$ with $H = \frac{|\mathbf{p}|^2}{2} - \frac{k}{r}$.

(a) **Sets 1–2**: The orbit of a planet is the sequence $(q(t), p(t))$ generated by the Hamiltonian flow. What structure from Sets 1–2 can already describe? What is the "f" here?

(b) **Sets 3–4**: The phase space $T^*(\mathbb{R}^3 \setminus \{0\})$ is a topological space. Is it compact? Is it Hausdorff? Does the Extreme Value Theorem apply to H? For bound orbits ($H < 0$): are the orbits in a compact subset?

(c) **Sets 5–6**: The flow is smooth. The velocity of the planet at each instant is $\dot{q}^i = \frac{\partial H}{\partial p_i}$ a tangent vector to the orbit. Which structure from Set 6 guarantees this is well-defined?

(d) **Sets 7**: The equations of motion are the integral curves of X_H on $T^*(\mathbb{R}^3 \setminus \{0\})$. From Set 7: the Noether theorem gives J_i conserved from rotation symmetry. Write the symmetry vector fields and the corresponding conserved quantities.

(e) **Sets 8–9**: The Liouville 1-form $\alpha = \sum p_i \, dq^i$ is a differential 1-form. Its integral $S = \int_\gamma \alpha$ over an orbit gives Hamilton's principal function the key to Hamilton-Jacobi theory.

(f) **Set 10**: The de Rham cohomology class $[\omega] \in H^2_{dR}(T^*(\mathbb{R}^3 \setminus \{0\}))$. Is ω exact? (Yes: $\omega = -d\alpha$.) What does this say about the Kepler problem?

(g) **Set 11**: The conserved angular momenta J_1, J_2, J_3 satisfy $\{J_i, J_j\} = -\epsilon_{ijk} J_k$ the Lie algebra $\mathfrak{so}(3)$ appearing as Poisson brackets. The symmetry group of the Kepler problem is $SO(3)$ (rotations).

(h) **Set 13**: The symplectic form ω gives the Poisson bracket $\{H, J_i\} = 0$, proving conservation. The Hamilton-Jacobi method gives the complete solution: elliptical orbits, $T^2 \propto a^3$.

(i) **The complete answer**: starting from a set and a function (Set 1), adding thirteen layers of structure, we can now:

- Prove the orbits exist and are continuous (topology)
- Prove fixed points exist (compactness + IVT)
- Write the equations of motion (smooth structure)
- Identify the symmetries and conservation laws (Lie groups + symplectic)
- Solve the equations completely (Hamilton-Jacobi)
- Derive the shape of orbits (Kepler's first law)
- Derive the period (Kepler's third law)

Write one sentence: what is the single most important structure that makes this possible, and why?

B4. (Physics detecting topology the complete thread.)

Throughout these sets, we have seen physics detect topology. Collect all examples:

Set	**Physics**	**Topology detected**	**Invariant**
4	Winding number	Hole in $\mathbb{R}^2 \setminus \{0\}$	$n \in \mathbb{Z}$
6B	Morse/SQFT	$\chi(M)$	$\sum(-1)^k \dim H^k$
6B	Gauss-Bonnet	Genus of surface	$2 - 2g$
8	Maxwell $dF = 0$	H^2 of spacetime	Magnetic flux
9	Gauss's law	Hole in $\mathbb{R}^3 \setminus \{0\}$	Charge Q
10	Witten index	$\chi(M)$	Ground states
11	Spin-$\frac{1}{2}$	$\pi_1(SO(3)) = \mathbb{Z}_2$	Spinor sign
12	Dirac monopole	$c_1 \in \mathbb{Z}$	Charge quantization
13	Kepler	$SO(3)$ symmetry	Elliptical orbits

(a) In each case, identify the common pattern: a physical quantity takes values in $\mathbb{Z}$ (or $\mathbb{Z}_2$) because of a topological invariant. Why must topological invariants be discrete?

(b) The deepest example is the Dirac monopole (Set 12): $c_1 \in \mathbb{Z}$ forces charge quantization. The deepest *open* question: does charge quantization in nature actually come from a monopole? What experiment would confirm this?

(c) **The grand conclusion.** Mathematics provides the language (sets, topology, manifolds, forms, Lie groups, bundles, symplectic geometry). Physics provides the questions (what is the motion? what is conserved? why is charge quantized? why does spin-$\frac{1}{2}$ exist?). The answers are always at the intersection. Write one paragraph describing what you think the next layer of structure beyond symplectic geometry should be, based on what physics still cannot explain in this framework.
(Possible answers: quantum mechanics replacing $\{f, g\}$ with $\frac{1}{i\hbar}[f, g]$; general relativity replacing the symplectic form with the spacetime metric; quantum field theory replacing finite-dimensional phase space with infinite-dimensional field space; string theory replacing point particles with 1*-dimensional objects.)*

Thirteen sets later, we have built the complete mathematical framework of classical physics: topology gave us convergence, smooth structure gave us velocity, differential forms gave us integration and gauge fields, Lie groups gave us symmetry and spin, fiber bundles gave us the Dirac monopole and charge quantization, and the symplectic form gave us Hamiltonian mechanics and Kepler's laws.

At every step, the new structure answered a question the previous one could not and raised a new question that the next one would answer.

The next questions physics asks why is energy quantized? why does the electron have a definite mass? what is the geometry of spacetime at small scales? will require new mathematics. Some of that mathematics does not yet exist. You now know enough to start looking for it.***Final Reflection.*** *We began with a set X and a function $f : X \to X$. We asked: what are the orbits?*

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